

The Cosmological Constant as a Sixth Power of the MOND Scale: A Transformation of the Fine-Tuning Problem

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Abstract

The cosmological constant problem—why Λ is 122 orders of magnitude smaller than the Planck scale prediction—is widely regarded as the deepest fine-tuning problem in physics. In a companion paper, we showed that the MOND acceleration constant satisfies $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$ to 1%. This relationship can be inverted: $\Omega_\Lambda = \Omega_m[2\pi a_0/(cH_0)]^6$. If a_0 is treated as the more fundamental quantity, the cosmological constant is not an independent parameter—it is determined by the acceleration at which gravity changes character. The fine-tuning problem transforms from “why is Λ so small?” to “why does gravity change character at a_0 ?” The second question, unlike the first, has a physical context (the galaxy rotation curve transition) and, in the causal diamond framework, a geometric answer (the waist of the observer’s causal diamond). We examine three interpretations of the Λ – a_0 relationship, their implications, and their limitations.

1. The Cosmological Constant Problem

Quantum field theory predicts a vacuum energy density of order $\rho_{\text{vac}} \sim m_{\text{Pl}}^4/\hbar^3 c^3 \sim 10^{113} \text{ J/m}^3$. The observed value, inferred from the accelerated expansion of the universe, is $\rho_\Lambda \sim 10^{-9} \text{ J/m}^3$. The discrepancy of ~ 122 orders of magnitude is the cosmological constant problem [1, 2].

No known mechanism explains this cancellation. Proposed approaches include anthropic selection [3], supersymmetric cancellation [4], and various dynamical relaxation mechanisms [5], but none has achieved consensus. The problem is widely regarded as a crisis for fundamental physics.

2. The Inversion

In Ref. [6], we showed empirically that the MOND acceleration constant [7] satisfies:

$$a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6} / (2\pi) \quad (1)$$

to 1% accuracy [6], with the geometric origin of the 2π and the matter–dark energy correction subsequently derived from the causal diamond waist [8, 9]. Equation (1) can be inverted algebraically:

$$\Omega_\Lambda = \Omega_m \times [2\pi a_0 / (cH_0)]^6 \quad (2)$$

Numerically, with $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$, $H_0 = 67.4 \text{ km/s/Mpc}$, and $\Omega_m = 0.315$: $[2\pi a_0/(cH_0)] = 1.152$, raised to the sixth power gives 2.33, so $\Omega_\Lambda = 0.315 \times 2.33 = 0.734$. The observed value is $\Omega_\Lambda = 0.685$. The agreement is 7%, reflecting the 1% residual in Eq. (1) amplified by the sixth power. The inversion is not exact, but it recovers the correct order of magnitude and the correct value to within $\sim 10\%$ —far better than the 122 orders of magnitude predicted by QFT.

3. Three Interpretations

3.1 Interpretation 1: Λ is fundamental, a_0 is derived

In the standard picture, Λ is a parameter of the gravitational Lagrangian and $a_0 = c^2/(2\pi r_w)$ is a derived quantity that follows from the causal diamond geometry [8]. Equation (2) is then a mathematical identity, not a physical inversion. The cosmological constant problem remains: why is Λ so small? The causal diamond framework explains what a_0 is, but does not explain why Λ has the value that produces it.

3.2 Interpretation 2: a_0 is fundamental, Λ is derived

If a_0 is instead treated as the fundamental quantity—an acceleration scale built into the laws of gravity or inertia—then Eq. (2) determines Λ . The cosmological constant problem transforms:

	Standard problem	Transformed problem
Question	Why is $\Lambda \sim 10^{-122} \text{ m}^{-2}$?	Why is $a \sim 10^{-10} \text{ m/s}^2$?
Fine-tuning	122 orders of magnitude	62 orders of magnitude
Physical context	None (Λ is a bare parameter)	Galaxy rotation curves transition here
Geometric answer	None	Causal diamond waist [8]

Table 1. The cosmological constant problem in standard and transformed form.

The sixth-power relationship means that a 1% change in a_0 produces a ~6% change in Ω_Λ . The fine-tuning is reduced from 122 to 62 orders of magnitude (in Planck units), but not eliminated. However, the transformed problem has something the standard problem lacks: a physical scale. The question “why is Λ small?” has no natural comparison point. The question “why does gravity change at a_0 ?” points directly to the galaxy rotation curve transition, the causal diamond waist, and the entropy balance between area-law and volume-law degrees of freedom.

3.3 Interpretation 3: Both are emergent

In the causal diamond framework [8, 9, 10], both Λ and a_0 are features of the observer’s causal structure. Λ determines the shape of the causal diamond (through the Friedmann equation). a_0 is the acceleration at the waist of that diamond. They are not independent quantities related by a formula—they are two aspects of the same geometric object. In this view, the expansion history—which is directly observable—is the ground truth, and both Λ and a_0 are descriptions of its geometry: Λ names its late-time asymptotic behavior, a_0 names the acceleration at the waist of its causal diamond.

In this interpretation, Eq. (2) is not an inversion but a consistency relation. The cosmological constant problem does not transform; it dissolves. The question “why is Λ so small?” becomes “why does the causal diamond have the shape it has?”—which is answered by the Friedmann equation with the observed matter and energy content. There is no fine-tuning because there is no free parameter: Λ and a_0 are both determined by the expansion history.

This interpretation is the most radical and the most speculative. It requires that the cosmological constant not be an independent input to the theory but rather an emergent property of the causal structure. Whether this is tenable depends on the resolution of the entropy normalization ambiguity discussed in Ref. [10] and on the development of a microscopic theory of the causal diamond.

4. Why the Sixth Power?

The exponent 6 in Eq. (2) is not arbitrary. In Ref. [6], it was shown empirically that the MOND acceleration constant scales as the 1/6 power of the dark energy to matter ratio. The geometric-mean bridge analysis of Ref. [11] connected this to the relationship $a_0 = (a_{\text{now}} \times a_{\text{eq}})^{1/2}$, where $a_{\text{now}} = cH_0$ is the present Hubble acceleration and a_{eq} is the acceleration at matter–dark energy equality.

In the causal diamond picture, the 1/6 power reflects the interplay between three spatial dimensions (r^3 in the volume-law entropy) and two surface dimensions (r^2 in the area-law entropy). The entropy crossover condition $S_{\text{area}} \sim S_{\text{vol}}$ equates a quadratic to a cubic, and the solution involves the ratio of scales raised to the power determined by the dimensionality difference. The precise origin of the 1/6 from the causal diamond geometry remains an open derivation; Ref. [6] demonstrated its empirical validity and Ref. [12] explored its algebraic structure.

5. Caveats

Several important limitations apply. First, the inversion Eq. (2) recovers Ω_Λ to ~10%, not exactly. The sixth power amplifies any residual in Eq. (1). This is not the precision needed to “solve” the cosmological constant problem; it is sufficient only to demonstrate that Λ and a_0 are related by a low-power algebraic formula rather than being independent.

Second, Interpretation 2 (a_0 fundamental) lacks a theoretical mechanism. No known framework makes a_0 a fundamental constant from which Λ can be derived. The MOND Lagrangians (AQUAL [13], TeVeS [14], AeST [15]) all treat a_0 as a parameter, not a prediction. The inversion is a mathematical observation, not a theoretical derivation.

Third, Interpretation 3 (both emergent) requires that the Friedmann equation itself be fundamental—that the expansion history is the ground truth from which all other quantities follow. This is a stronger claim than the causal diamond framework has established.

This paper identifies a mathematical relationship and explores its interpretations. It does not solve the cosmological constant problem. It shows that the problem can be reframed in a way that connects it to observable physics (the MOND transition) and to a specific geometric structure (the causal diamond waist).

6. Conclusion

The relationship $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$, validated to 1% in Ref. [6] and geometrically grounded in Refs. [8, 9], can be inverted to express the cosmological constant in terms of the MOND acceleration scale. The cosmological constant problem—“why is Λ so small?”—transforms into “why does gravity change character at a_0 ?”

The second question has a physical context (galaxy dynamics), a geometric answer (the causal diamond waist), and a falsifiable prediction (the redshift evolution of a_0 [16]). Whether this transformation represents a genuine reduction of the cosmological constant problem or merely a restatement depends on which quantity is more fundamental—a question that cannot be settled by the present analysis but that connects one of the deepest problems in theoretical physics to one of the most well-measured phenomena in observational astrophysics.

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